Assignment No.6 2413FEB2T1 : Engineering Mathematics-2
Topic: Multiple Integrals

1 Problems on Double Integration

1. Evaluate $\int_0^1 \int_x^{x^2} xy(x^2+y^2)dydx$.
2. Evaluate $\int_0^1 \int_0^y xye^{x^2}dxdy$.
3. Evaluate $\int_0^1 \int_0^{x^2} e^{\frac{y}{x}}dxdy$.
4. Evaluate $\int_0^1 \int_y^{10y} \sqrt{xy-y^2}dxdy$.

2 Problems on Region with given boundaries

1. Evaluate $\iint_R ydxdy$ where R is the region bounded by the line $y = x$ and the parabola $y = 4x - x^2$.
2. Evaluate $\iint_R xydxdy$ where R is the region bounded by $y = x^2$ and the parabola $y^2 = -x$.
3. Evaluate $\iint_R x^2dxdy$ where R is the region bounded by the curves $xy = 16$, $x = y$, $y = 0$ and $x = 8$ in first quadrant.

3 Problems on Change of Order

1. Change the order of $\int_0^5 \int_{2-x}^{2+x} f(x,y)dxdy$.
2. Change the order of $\int_0^a \int_{\sqrt{a^2-x^2}}^{x+3a} f(x,y)dxdy$
3. Change the order and evaluate $\int_0^2 \int_{\frac{x^2}{2}}^{4-x} xydxdy$

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4 Problems on Polar Coordinate system

1. Evaluate $\iint_R x^2y^2dxdy$ where R is the region enclosed by ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.
2. Evaluate $\iint_R y^2dxdy$ where R is the region enclosed by $x^2+y^2-ax=0$ and $x^2+y^2-2ax=0$.
3. Evaluate $\iint_R \frac{dxdy}{(1+x^2+y^2)^2}$ where R is the region bounded by one loop of $(x^2+y^2)^2=x^2-y^2$.

5 Problems on Area using double Integration

1. Find the area bounded by $y^2=4x$ and $2x-3y+4=0$.
2. Find the area of cardiode $r=a(1+cos\theta)$.
3. Find the area common to the circles $r=2cos\theta$ and $r=2sin\theta$.
4. Find the area bounded by the circle $x^2+y^2=4x$.

6 Problems on Triple Integration

1. Evaluate $\int_0^{log2}\int_0^x\int_0^{x+y} e^{x+y+z}dxdydz$
2. Evaluate $\int_0^1\int_{y^2}^1\int_0^{1-x} xdxdydz$
3. Evaluate $\int_0^1\int_0^{1-x}\int_0^{1-x-y} \frac{dxdydz}{(1+x+y+z)^3}$

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